

SIROP: High-Level Synthesis with Parallel Patterns

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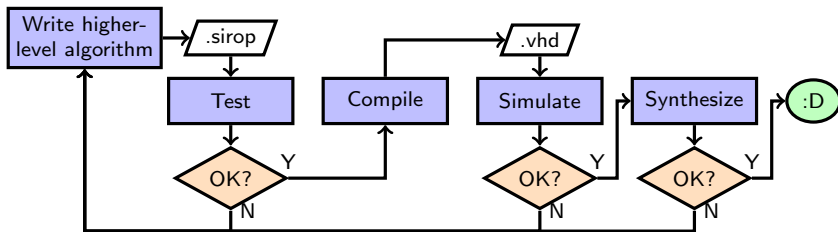


McGill

September 22, 2026

Recap: Project Goal

Goal: simplify the implementation of resource-efficient data processing (e.g., energy reconstruction algorithms) on FPGA



Latency matching



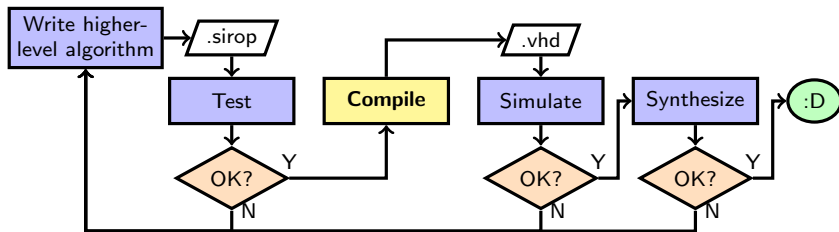
Fusion



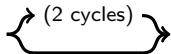
etc.

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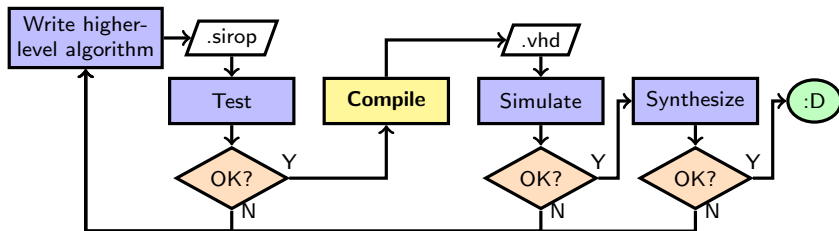
Fusion



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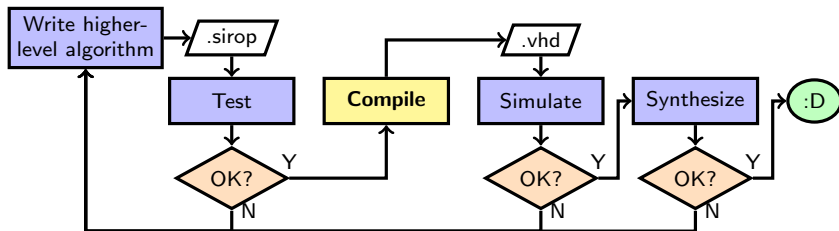
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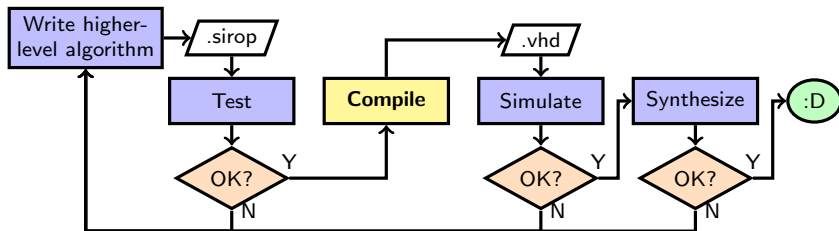
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etc.

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etc.

Previous Results

6-tap filter with hard-coded weights, targeting Agilex 7

Design	$f_{\max}$	ALMs	DSPs
Handwritten VHDL	880 MHz	115*	3
Generated w/ SIROP	510 MHz	98* (-15%)	3

*Excluding ALMs reported as “unavailable due to virtual I/Os.”

Why the Difference in ALM Usage?

1 Inconsistent compilation settings

- I didn't notice the device part number for McGill's devkit differs from the `G_BSP_AGI_NO_HW` target used in LASP-dacore :(
(

2 Different DSP pipelining settings

- See merge request [!143](#) in LASP-dacore
- See presentation explaining DSP register packing:
<https://indico.cern.ch/event/1700332/>

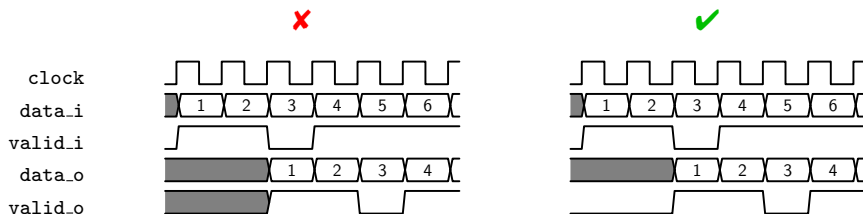
Updated Results (!!! ANTICIPATED !!!)

6-tap filter with hard-coded weights, targeting Agilex 7

Design	$f_{\max}$	ALMs	DSPs
Handwritten VHDL*	810 MHz	118 [†]	3
Generated w/ SIROP	810 MHz	118 [†]	3

* After merge request [!143](#)

Initial valid Outputs



- It's good to ensure the `valid` field in `t_avst_stream` outputs is 0, even immediately after reset
 - See merge request [!142](#) in LASP-dacore
- SIROP can check such conditions (e.g., `valid` bit must be 0 for the first n cycles, where n is the entity's latency)

DSP Input Cascade

- TODO: explain FIR filter trick about selecting every 7th rather than every 8th index in shift register

SIROP can automatically transform the simpler version into this resource-efficient version (by merging and shrinking redundant shift registers)!

Conclusion

- SIROP compiler matches resource efficiency of handwritten FIR filter, from much more concise source code
- Next steps: more advanced energy reconstruction algorithms
 - Neural networks
 - Optimal filter with forward corrections



(SIROP repository)